SPILL: An anonymous peer-listener web Platform

Description of the concept of the Mission

Roughly one psychiatrist serves every million people in sub-Saharan Africa, and Rwanda reflects that shortage with about 18 psychiatrists nationwide for a population with a median age of 19.9 (UNICEF & World Health Organization [WHO], 2023; University of Global Health Equity, 2026; DataReportal, 2025), while stigma keeps many young people from the services that do exist (UNICEF Rwanda, 2020). My mission is to tackle mental health stigma and emotional isolation among African youth by creating a safe space where they can express themselves and connect with others who understand. Spill is the first step towards that mission: a free, English-language web platform giving young Rwandans aged 16 to 25 a private, anonymous way to talk to someone who understands, piloted in Rwanda within the wider African youth population it addresses. The casual, non-clinical name is deliberate, since stigma is a leading reason young Rwandans avoid help (UNICEF Rwanda, 2020), and the approach, task-shifting first-line support to trained non-specialists, mirrors the Friendship Bench trial in Zimbabwe, which reduced depression symptoms in person among adults (13.7% versus 49.9% with usual care) (Chibanda et al., 2016). The timing fits a documented gap between awareness (61.7%) and service use (5.3%) (Kayiteshonga et al., 2022) and growing community demand, such as The Circle Kigali's 89 circles and over 3,000 participants in two years (The Circle Kigali, n.d.-b).

Problem statement

The 2018 Rwanda Mental Health Survey (n = 19,110) found 20.49% of people had one or more mental disorders, rising among young people from 10.2% at ages 14 to 18 to 17.2% at 19 to 25, and more recent studies found rates as high as 14.48% of 5,221 youth screening positive, depression (51%) and anxiety (67.3%) symptoms among 384 secondary students, and past-month suicidal thoughts in up to 21% of young adults (Kayiteshonga et al., 2022; UNICEF Rwanda, 2020; Ndikumana et al., 2025; Mukantwali et al., 2024; Muwonge et al., 2019). Yet only 5.3% of Rwandans have used a mental health service despite 61.7% awareness, and only 38.3% know where to seek support, largely because of stigma, cost, not recognising it as a problem, and beliefs attributing it to spiritual causes (Kayiteshonga et al., 2022; UNICEF Rwanda, 2020).

Young Rwandans aged 16 to 25, mostly connected urban students and early-career youth, feel this gap hardest: hospital services and helplines are formal and voice-only, digital options are information-only or limited to university students, and in-person therapy costs 47,200 RWF a session (Kigali Mental Health, n.d.; Sezibera et al., 2025; The Circle Kigali, n.d.-c), while distress itself does not follow office hours. The problem is national, though the first pilot is Kigali-based; with only 34.2% internet use against 92% mobile connections (DataReportal, 2025), Spill targets connected youth first with anonymous listener chat, a mood check-in, verified local resources, and message screening, so that risk is managed while users only ever reach vetted listeners, never each other.

Gap and uniqueness

To the best of my knowledge, from searches carried out in September 2026 and to be reconfirmed before final submission, no Rwandan platform offers anonymous, on-demand, one-to-one peer-listener chat for young people. Kigali Mental Health and the national helplines (114, 116, 8015) exist but stay clinical, in-person, or voice-only, with no chat and no follow-up (Kigali Mental Health, n.d.; Rwanda Biomedical Centre [RBC], n.d.; Africa Press, 2021); E-Ruhuka's 2025 pilot reached only university students and reported no anonymous listener chat (Sezibera et al., 2025); CyberRwanda serves sexual and reproductive health, not emotional support (YLabs, n.d.); The Circle Kigali is in-person, named, and mostly paid (The Circle Kigali, n.d.-a, n.d.-c); and 7 Cups offers anonymous peer chat internationally but isn't localised to Rwandan hotlines, clinics, or law (7 Cups, n.d.). Spill is the first to combine all four: anonymous, on demand, chat-based, and tied to local referral, built safety-first (new users reach only vetted listeners, with no public feeds or user-to-user messages), Rwanda-specific (crisis language triggers verified local hotlines, and mood ratings give measurable outcomes), and right-sized as a free, lightweight web app for low-cost devices, designed around Rwandan law (Republic of Rwanda, 2021).

Software development model

Selected model: Agile (Scrum-based). Requirements kept shifting through research, the data protection law pushed the minimum age to 16, and safety features like message screening need testing early and often, not once at the end, so Agile's two-week sprints suit a solo developer better than a plan-first model over a fixed 8-week semester. Roles are combined: I act as product owner and developer, with my lecturer and a counsellor adviser as stakeholder reviewers. Each sprint follows the same cycle:

1**. Product backlog**: User stories with acceptance criteria go into GitHub Issues, prioritised Must, Should, Could, Won't, covering features such as pseudonymous registration, the resources page, mood check-in, listener chat and message screening.

2. **Sprint planning**: I pull the highest-priority stories into each two-week sprint, following the plan below, for example the screening module and moderator queue in Sprint 3.

3. **Build**: Daily written logs (done, next, blocked) replace stand-ups; work happens on feature branches in GitHub.

4. **CI and testing**: GitHub Actions runs automated tests on every push, especially on the message-screening module.

5. **Sprint review**: I demo the increment against the Definition of Done: tested, reviewed, documented, deployed.

6. **Retrospective**: I note what to keep, change or stop, and adjust the backlog.

7. **Release**: Each sprint deploys to a hosted test site; a small pilot follows Sprint 4.

Sprint	Weeks	Goals	Deliverable
1	1-2	Foundation	Repo/CI setup, pseudonym login,resources page, mood check-ins
2	3-4	Talk to a listener	Listener matching queue, real-time chat (Firestore)
3	5-6	Make it safe	Rule-based message screening (Firestore Security Rules), report/block, moderator queue
4	7-8	Prove and ship	Testing, deployment, demo

The hypothesis of the solution

If Spill launches, I expect young Rwandans who would not otherwise seek help to use it as a first, private step (at least half of pilot users, by onboarding survey), most to feel a little better right after a listener chat (mood check-in before and after, holding for at least 60% of chats), the screening module to reliably catch crisis language and contact-sharing attempts (at least 90% of each in a test set), and enough people to return that they would use Spill again (at least 70%, by post-chat survey). These are pilot goals I have set myself, tested in the class pilot; the null hypothesis is that mood ratings show no real difference before and after. The pilot itself uses consenting adults 18 and older, or a supervised partner organisation, since minors' consent under Rwandan law needs confirming first. Spill offers peer support, not therapy.

References

Africa Press. (2021). *New suicide prevention helpline: How it works*. Africa press. https://www.africa-press.net/rwanda/all-news/new-suicide-prevention-helpline-how-it-works

Chibanda, D., Weiss, A. H., Verhey, R., Simms, V., & Munjoma, R. (2016). Effect of a primary care–based psychological intervention on symptoms of common mental disorders in Zimbabwe: A randomized clinical trial. *JAMA*, *316*(24), 2618–2626. https://doi.org/10.1001/jama.2016.19102

The Circle Kigali. (n.d). *The Circle Kigali*. Events. https://thecircle.sinc.events/

The circle Kigali. (n.d). *The Circle Kigali*. Services. Retrieved September 21, 2026, from https://thecirclekigali.com/services

The Circle Kigali. (n.d). *The Circle Kigali*. posts. https://www.instagram.com/thecircle.kigali/

CyberRwanda — YLabs. (n.d). YLabs. Retrieved September 27, 2026, from https://www.ylabsglobal.org/work/all/cyberrwanda

Data Reportal. (2025). *Digital 2025: Rwanda*. https://datareportal.com/reports/digital-2025-rwanda

Kayiteshonga, Y., Sezibera, V., Mugabo, L., & Iyamuremye, J. D. (2022). Prevalence of mental disorders, associated co-morbidities, health care knowledge and service utilization in Rwanda towards a blueprint for promoting mental health care services in low- and middle-income countries? *BMC Public Health*, *22*, 1858. https://doi.org/10.1186/s12889-022-14165-x

Kigali Mental Health. (n.d.). *Kigali Mental Health*. https://www.kmentalhealth.gov.rw/

Law relating to the Protection of Personal Data and Privacy. (n.d.). RwandaLII. Retrieved September 27, 2026, from https://rwandalii.org/akn/rw/act/law/2021/58/eng@2021-10-15

Mukantwali, M. B., Niyonsenga, J., Uwingeneye, L., Kanyamanza, C. U., & Mutabaruka, J. (2024). Prevalence of depression, anxiety and stress and their risk and protective factors among secondary students in Rwanda during the first wave of COVID-19 pandemic. *PLOS ONE*, *19*(7), e0306389. https://doi.org/10.1371/journal.pone.0306389

Muwonge, J., Umubyeyi, A., Rugema, L., & Krantz, G. (2019). Suicidal behaviour and clinical correlates in young adults in Rwanda: A population-based, cross-sectional study. *Journal of Global Health Reports*, *3*, e2019080. https://doi.org/10.29392/joghr.3.e2019080

Ndikumana, F., Izabayo, J., Kalisa, J., Nemerimana, M., & Nyabyenda, E. C. (2025). Machine learning-based predictive modelling of mental health in Rwandan youth. *Scientific Reports*, *15*, 16032 (article no.). https://doi.org/10.1038/s41598-025-00519-z

Rwanda Biomedical Centre. (n.d). *Locate Our services*. https://rbc.gov.rw/index.php?L=10&id=631

Rwanda Public Health Bulletin. (2025). *Assessing the determinants of mental health services utilization among adolescents aged 18–24 years in Kicukiro district, Rwanda, 2025*. Rwanda Biomedical centre. https://rbc.gov.rw/publichealthbulletin/articles/read/182/

7 cups. (n.d). *7 cups*. wikipedia. https://en.wikipedia.org/wiki/7_Cups

Sezibera, V., Twizere, C., & Izabayo, J. (2025). Evaluating the usability, user experience, and satisfaction of the E-Ruhuka mental health mobile application among university students in Rwanda, 2025. *Rwanda Public Health Bulletin*, *6*(2), 7-15. https://doi.org/10.4314/rphb.v6i2.2

UNICEF Rwanda. (2020). *Adolescent mental health landscape assessment in Rwanda: Final report*. UNICEF Rwanda. https://www.unicef.org/rwanda/media/39916/file/AMH%20Landscape%20Assessment%20Rwa

da.pdf

UNICEF & World Health Organization. (2023). *Mental health is a human right, but only 1 psychiatrist per 1000000 people in sub-Saharan Africa – UNICEF/WHO*. UNICEF. Retrieved September 27, 2026, from https://www.unicef.org/esa/press-releases/mental-health-a-human-right

University Of Global Health Equity. (2026). *Psychiatry, Reimagined for Equity – University of Global Health Equity*. University of Global Health Equity. Retrieved September 27, 2026, from, https://ughe.org/psychiatry-reimagined-for-equity/

Software Requirements Specification (SRS)

Title: **SPILL**

Prepared by: **Dianah Teta**

Organization: **African Leadership University**

Date Created: **21 September 2026**

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Revision History

Name	Date	Reason For Changes	Version
Dianah Teta	29 September 2026	Initial version of the project proposal and SRS specification for SPILL	1.0
Dianah Teta	26 October 2026	Added appendix B with UML diagrams	2.0

1. Introduction

1.1 Purpose

This document specifies the software requirements for **Spill, version 1.0 (minimum viable product)**, an anonymous peer-listener web platform for young people aged 16 to 25 in Rwanda. It covers the complete product: the public web application, the listener and moderator tools, and the administration pages. The following are outside the scope of version 1.0 and are listed as future work: group community spaces, users under 16, machine-learning moderation, a native mobile app, phone verification, and Kinyarwanda or French interfaces.

1.2 Document Conventions

- Requirement identifiers use the prefixes **FR** (functional), **NFR** (non-functional) and **BR** (business rule), for example FR 4.2.
- Statements use “shall” for mandatory behaviour. Every requirement carries its own priority, shown in brackets at the end of its description using MoSCoW: **Must, Should, Could**.
- Bold marks requirement identifies and defines roles. Terms defined in Appendix A appear with an initial capital. “Listener” means a vetted volunteer, and “User” means a young person seeking support.
- Access levels: **Tier 0** is open to anyone without an account; **Tier 1** is a registered pseudonymous account. Tier 2 (organisation-coded group spaces) is out of scope for version 1.0.

1.3 Intended Audience and Reading Suggestions

Reader	Interest	Suggested section
Facilitators and assessors	Overall product, Feasibility and quality of requirements	1,2,4
Developer(author)	Detailed behaviour to build and test	3,4,5
Testers and peer reviewers	Behaviour to verify, acceptance checks	4,5
Counsellor advisors and moderators	Safety, Escalation and resources	2.3(FR 6),5
Investors	Scope, Privacy and Legal Position	1.4,2.5

1.4 Product Scope

Spill is a web app with three parts: an open resources area (verified helplines, clinics and self-help content, no account needed), a pseudonymous area for mood check-ins and one-to-one listener chat, and a staff area for listeners, moderators and administrators. Every message is screened, and users can reach only vetted listeners, never each other. The goal is a private, stigma-free first step and a route to professional care when needed, measured against four objectives for version 1.0: register and start a chat in under five minutes (O1); screen every message for crisis language and contact-sharing (O2); moderator review of every flag (O3); and before-and-after mood ratings on every chat (O4).

This responds to Rwanda's push to expand mental health access and reduce stigma (allAfrica, 2024), and the gap between 61.7% awareness and 5.3% service use found nationally (Kayiteshonga et al., 2022). Of Rwanda's roughly 2 million 18–24 year-olds, only 34.2% are online (DataReportal, 2025); existing spaces such as The Circle Kigali, with 89 circles and over 3,000 participants in two years (The Circle Kigali, n.d.-b), point to real demand. The pilot target is therefore modest: about 30 users and 5 trained listeners.

1.5 References

allAfrica. (2024, September 12). *Rwanda steps up mental health efforts to combat suicide.* https://allafrica.com/stories/202409120046.html

Sources already listed in the proposal's References (Part 1) are not repeated here, even where the SRS cites them again.

2. Overall Description

2.1 Product Perspective

Spill is a new, self-contained product. It does not replace or connect to any existing system, and it links out to public services such as hotlines and clinics through an admin-maintained directory, rather than integrating with them technically; there is no live data exchange with those services, only a curated list an administrator keeps current. Three types of user interface with it directly, each over a secure (HTTPS) connection: young people aged 16 to 25 on a phone or laptop, vetted listener volunteers, and moderators and administrators. All three reach the system through the same web front end; version 1.0 has no separate mobile app or standalone admin tool. What each of these users can do within the product is set out in Section 2.2, and any future integration with national hotline systems or a native mobile app is treated as later work, outside this version's scope (Section 1.1).

2.2 Product Functions

The high-level functions are listed below. Detailed requirements are in Section 4.

- **Access and accounts**: register with a pseudonym and age band, sign in and out, delete an account.
- **Open resources (Tier 0)**: verified helplines, clinic directory and self-help content, plus a “Need help now” button on every page.
- **Mood check-in**: record mood, view history, and see a private monthly recap (Could).
- **Listener chat**: request a chat by topic, be matched with an approved listener, exchange messages, end the chat, and rate it.
- **Listener management**: apply, be vetted and approved, complete training, set availability, and accept chats.
- **Safety and moderation**: screen every message, show crisis support, block contact-sharing, report and block, and review flags in a moderator queue with an audit trail.
- **Administration**: manage keyword lists, resources, roles and anonymised statistics.

2.3 User Classes and Characteristics

User Class	Characteristics	Privileges and use	Priority
Visitor(Tier 0)	Anyone, no account; may be in distress and want no data collected	View helplines, clinics and self-help content only	High
User(Tier 1)	Aged 16–25, English-reading, on a phone or laptop; two age bands: 16–17 and 18–25; low tolerance for friction or judgement	Mood check-in; request chats with listeners; report and block; delete own data	Most important
Listener	Vetted volunteer, 21 or older, trained in active listening; only listeners cleared for minors may be matched with the 16–17 band	Accept user-started chats; cannot initiate contact; strict message checks; own dashboard	Most important
Moderator	Trained staff or volunteer with safeguarding responsibilities	Review flags and reports, warn or suspend accounts; access chat content only through a reviewed case	High
Administrator	System owner (developer); technical	Approve listeners, manage lists and	Medium

		resources, view anonymised statistics; cannot read chats outside the moderation process; all actions logged	

2.4 Operating Environment

- **Client**: browser-only responsive web app, no install; current versions of Chrome, Edge, Firefox and Safari on Android, iOS, Windows, macOS and Linux; screen widths from 320 px upwards
- **Server:** Firebase e Hosting (free Spark plan) serves the web app; there's no separate application server; the client talks directly to Firestore and Authentication.
- **Database**: Cloud Firestore (NoSQL), used for both development and the hosted pilot, secured by Firestore Security Rules.
- **Coexistence**: no other software on the client is required; the app must work without browser extensions

2.5 Design and Implementation Constraints

- **Legal:** Law No. 058/2021 requires guardian consent for a child under 16 (Art. 9), classes health status as sensitive data (Art. 3), requires registration with the National Cybersecurity Authority (Art. 29), a 48-hour breach notification (Art. 43), a data protection officer for large-scale sensitive data (Art. 40), and storage of personal data in Rwanda unless the registration certificate authorises storage or transfer abroad (Arts. 48 and 50) (Republic of Rwanda, 2021). Version 1.0 therefore excludes under-16s.
- **Resources**: one developer, 8 weeks, no budget; the entire stack runs on Firebase's free Spark Plan (Authentication, Firestore, Hosting), with no billing account required; Spark caps daily reads, writes and storage rather than sleeping the server, which is generous enough for a ~30-user pilot but must be watched during testing. Firebase's servers are outside Rwanda, so the prototype uses test data and consenting adult volunteers only; real-user data would need Rwanda-based storage or authorisation (Arts. 48 and 50)
- **Technology:** Firebase (Authentication, Firestore, Hosting) on the free Spark plan, chosen for the developer's existing skills; chat uses Firestore's real-time listeners rather than polling or WebSockets.
- **Moderation**: rule-based screening only (no machine learning), enforced through Firestore Security Rules; human review is the safety net. A modified client could skip recording a blocked Attempt, but the message is still not delivered
- **Language**: English only, which limits detection of Kinyarwanda or French crisis wording.
- **Security and privacy**: no real names, phone numbers or location are collected; passwords are hashed; roles restrict access to chat content.

- **Ethics:** peer support is not therapy. The class pilot uses adults (18+) or a supervised partner pilot.

2.6 User Documentation

- In-app onboarding and a Help and FAQ page written in plain language.
- Privacy notice, terms of use, and confidentiality statement, including its limits when there is serious risk of harm.
- Listener training module with a short quiz, and a listener code of conduct.
- Moderator and administrator guide, and a README on GitHub covering set-up, testing and deployment

2.7 Assumptions and Dependencies

- Users have a smartphone or computer and internet access (the pilot targets connected youth).
- Enough volunteers can be recruited and vetted; police clearance is checked outside the system and recorded by the administrator.
- Hotline and clinic details remain valid. Each entry carries a “last verified” date and will be rechecked before release.
- Free-tier hosting remains available and adequate for about 30 users, and is used only with test data or consenting adult volunteers (Section 2.5).
- Minors’ consent rules for mental health support are confirmed with a legal or ethics adviser before any under-18 pilot.
- Community-engagement figures (for example, from The Circle Kigali) are indicative only.

3. External Interface Requirements

3.1 User Interfaces

The interface is responsive and mobile-first, with a calm colour palette and plain, non-clinical wording (for example “Rough day”, “Money stress”, “Study overload”). Standards that apply to every screen:

- A fixed “Need help now” button that opens the crisis card (verified helplines and clinics) in oneTap.
- Error messages are short, plain and never blaming; sign-in errors do not reveal whether an email exists

Screen	Purpose
Home	Explains the service; helplines, clinics and self-help; sign-up call to action
Sign up and sign in	Pseudonym, email, password, age band, consent and privacy notice

Dashboard	Mood check-in, history, start a chat, monthly recap
Chat	Message list, input, end chat, report, block, crisis card trigger
Listener application and training	Application form, training module and quiz
Listener dashboard	Availability toggle, incoming chats, training status
Moderator queue	Flags and reports ordered by severity; case view; actions
Admin pages	Approve listeners, manage keyword lists and resources, view statistics and audit log

3.2 Hardware Interfaces

Spill has no direct hardware interfaces. It runs in the browser of the user's own smartphone or computer and on a cloud-hosted server. No sensors, cameras, microphones or peripherals are used or required.

3.3 Software Interfaces

Exact SDK versions will be fixed in the project's package file in Sprint 1.

Component	Purpose
HTML, CSS and javascript via firebase hosting	Client-facing pages, connected live to Firestore
Cloud firestore	Storage of accounts, moods, chats, messages, flags, reports, resources and audit log
Firebase Authentication	Account creation, sign-in, pseudonymous identity, password handling
Firestore security rules	Access control, abuse limits, and rule-based message screening (crisis-language and contact-sharing patterns rejected at write time)
Jest and the Firebase local Emulator suite; ESLint	Automated testing, including rules testing, and code quality
Github and Github actions	Source control, backlog (Issues), and continuous integration

Firebase Authentication email verification (optional)	Account email verification (could)

Data items in and out: registration data (pseudonym, email, age band); mood entries; chat messages; Flags and reports; resource records; audit entries. Data is shared between components only through Firestore, with access restricted by Security Rules rather than application-level parameterised queries

3.4 Communications Interfaces

- All traffic uses HTTPS (TLS 1.2 or higher), enforced by default on Firebase Hosting and its services.
- The client stays in sync with chats, moods and flags in real time through the Firestore SDK over a persistent secure connection — no manual polling or custom REST endpoints.
- Firestore Security Rules screen every message at write time using regex-based checks (contact-sharing is rejected; crisis language is accepted only with its flag), and restrict who can Read or write each document, with public read-only access limited to Tier 0 content; Firebase Authentication guards sign-in against abuse
- There are no third-party push, SMS or message-broker services in version 1.0. Email is optional and used only for verification.

4. Requirement Specification

4.1 Stakeholder Requirements Specification

Stakeholder	Requirement
Young people(Users)	A private, quick and respectful way to talk to someone; no judgement, no exposure to strangers; clear help when things are serious; free of charge.
Listeners	Clear rules, training and protection; the ability to end unsafe chats; support from moderators
Moderators	A prioritised, manageable queue of flags and reports; the context needed to decide; a record of decisions.
Administrator/ developer	Control over listeners, lists and resources, but with logged, limited access to private content.
Lecturer and assessors	Traceable, testable requirements that follow

	the template and fit an 8-week project.
Investors (future)	Evidence of safeguarding, privacy and legal compliance before any pilot with minors.
Health System (indirect)	Consistent, verified referral information that points young people to hotlines and clinics

4.2 Functional Requirements

Priorities are shown at the end of each description

Req ID	Requirements	Description
FR 1	Access and accounts	
FR 1.1	Register Account	Allow a person aged 16 or older to register with an email, password and age band (16–17 or 18–25). Registration shall be refused for a declared age under 16. (Must)
FR 1.2	Assign pseudonym	Assign a random display name that is used everywhere instead of any real identity. (Must)
FR 1.3	Capture consent	Show the privacy notice, the limits of confidentiality and the peer-support (not therapy) notice, and require acceptance before the account is created. (Must)
FR 1.4	Sign in and out	Authenticate with email and password; end the session on sign-out or after inactivity. (Must)
FR 1.5	Delete account	Allow a user to delete their account and personal data; data tied to an open safety case is kept until the case closes. (Should)
FR 1.6	First time help seeking Question	Ask once at onboarding whether this is the first time the user has sought Support, to measure hypothesis H1. (Should)
FR 1.7	Verify email	Send a verification link to reduce fake and repeat accounts. (Could)
FR 2	Open Resources(Tier 0)	
FR 2.1	Display helplines	Show hotlines with number, purpose, language and a “last verified” date, without requiring an account. (Must)

FR 2.2	Display clinics and hospitals	Show a directory of mental health facilities with contact details and how to reach them. (Must)
FR 2.3	Display self-help content	Show short guides such as breathing, grounding and study-stress tips. (Should)
FR 2.4	Need help now button	Show a fixed button on every page that opens the crisis card in one tap. (Must)
FR 3	Mood check-in	
FR 3.1	Record mood	Let a user record mood on a 1–5 scale with emoji, optional topic tags and an optional short note. (Must)
FR 3.2	Record mood before and after chat	Capture a mood rating before a chat starts and after it ends and store both with the chat. (Must)
FR 3.3	View mood history	Show the user's own mood history as a list and simple chart. (Should)
FR 3.4	Private monthly recap	Show a private monthly summary; sharing is off by default. (Could)
FR 4	Listener Chat	
FR 4.1	Request a chat	Let a user choose a topic (for example exams, family, money, loneliness) and request a chat. (Must)
FR 4.2	Match with a listener	Place the request in a queue and match it with an available approved listener; users aged 16–17 are matched only with listeners cleared for minors. If none is available, show the wait status and Tier 0 resources. (Must)
FR 4.3	Exchange Messages	Send and receive text messages in real time via Firestore listeners. (Must)
FR 4.4	User initiated chats only	Only a user may start a chat; a listener shall not be able to initiate contact. (Must)
FR 4.5	End chat	Allow either participant to end a chat at any time without explanation. (Must)
FR 4.6	Rate chat	Offer an optional helpful / not helpful rating and short comment after a chat. (Should)
FR 4.7	Restrict Chat Visibility	Make chat content visible only to its two

		participants and, through a reviewed case, to moderators. (Must)
FR 5	Listener management	
FR 5.1	Apply as listeners	Provide an application form for volunteers, including the details needed for offline vetting. (Must)
FR 5.2	Approve listeners	Let the administrator approve or reject listeners after offline checks (identity, references, police clearance) and mark those cleared for minors. (Must)
FR 5.3	Training and quiz	Require a listener to complete a training module and pass a quiz before accepting chats. (Should)
FR 5.4	Set availability	Let a listener mark themselves available or unavailable. (Must)
FR 5.5	Approve or decline chats	Let a listener accept or decline a matched request, with a limit of two concurrent chats. (Should)
FR 5.6	Listener dashboard	Show a listener their own activity summary. (Could)
FR 6	Safety and moderation	
FR 6.1	Screen every message	Normalise each message (lower case, accents, repeated letters, common substitutions) and check it against pattern rules and term lists before delivery. (Must)
FR 6.2	Handle crisis language	On crisis wording, never block the sender: deliver the message, show the crisis card, and create a high-priority flag in the same atomic write. Security rules refuse a crisis message that has no matching flag. If the message also contains contact details, FR 6.3 applies: the message is rejected, but the crisis flag and card are still produced. (Must)
FR 6.3	Block contact sharing	Reject messages containing phone numbers, emails, links, social handles or requests to move off the platform, and warn the sender. The client then saves a separate blocked-attempt flag for the moderator queue (a listener sender is flagged for review), because Security Rules can only allow or deny a write. If a rejected message also contains

		crisis wording, the client also saves a high-priority crisis flag and shows the crisis card, so the sender still gets help. (Must)
FR 6.4	Flag abusive language	Flag insults and threats for moderator review; the recipient may end the chat. (Should)
FR 6.5	Report	Let any participant report a chat or the other participant with a reason. (Must)
FR 6.6	Block a person	Let a user or listener block the other party so they are not matched again. (Must)
FR 6.7	Moderator queue	Show flags and reports ordered by severity and age. (Must)
FR 6.8	Moderator actions	Let a moderator dismiss, warn, suspend, reinstate or escalate, with a recorded reason. (Must)
FR 6.9	Audit log	Record every moderator and administrator action and every access to chat content. (Should)
FR 6.10	Peer support notice	Show on chat screens that the service is peer support, not therapy or an emergency service. (Must)
FR 7	Administration	
FR 7.1	Manage term lists	Let the administrator edit crisis, contact-sharing and abuse lists without redeploying. (Should)
FR 7.2	Manage resources	Let the administrator add, edit and remove hotlines and clinics and set the last-verified date. (Must)
FR 7.3	View anonymised statistics	Show counts of chats, average mood change and flags, without personal data. (Should)
FR 7.4	Role based access	Enforce access by role (visitor, user, listener, moderator, administrator) through Firestore Security Rules; a user shall be able to read only their own chats. (Must)

5. Other Nonfunctional Requirements

5.1 Performance requirements

Requirement type	Req ID	Description
Performance	NFR 1	A page shall load in 3 seconds or less on a typical mobile connection. Rationale: young users abandon slow pages, and slow loading in a moment of distress is harmful. (Should)
performance	NFR 2	A sent message shall appear to the other participant within 2 seconds via Firestore's real-time sync. (Must)
Scalability	NFR 3	The system shall support at least 30 concurrent users during the pilot, within Firebase's free Spark plan quotas. (Should)

5.2 Safety Requirements

Requirement type	Req ID	Description
Safety	NFR 4	Crisis flags shall reach the moderator queue within 1 minute; the pilot target for moderator review is 24 hours. (Must)
Safety	NFR 5	The crisis card (verified helplines and clinics) shall be reachable in one tap from any page. (Must)

5.3 Security requirements

Requirement type	Req ID	Description
Security	NFR 6	Passwords shall be handled and hashed by Firebase Authentication and be at least 8 characters; the application shall never store or see plain-text passwords. (Must)
Security	NFR 7	All traffic shall use HTTPS (TLS 1.2 or higher), enforced by default on Firebase. (Must)
Security	NFR 8	Sessions shall use Firebase Authentication ID tokens, refreshed automatically and revoked on sign-out; every request for account, mood, chat, flag or staff data shall carry a valid token. The only exception is Tier 0 content (helplines, clinics, Self-help, crisis card), which is public and read-only (BR 8). (Must)

Security	NFR 9	Firestore access shall go only through Security Rules and the client SDK, never raw query strings, and all rendered output shall be escaped to prevent cross-site scripting. (Must)
Security	NFR 10	Sign-in attempts shall be limited by Firebase Authentication's built-in abuse protection; message sending shall be throttled by Firestore Security Rules (for example, a minimum interval between writes). (Must)
Privacy	NFR 11	The system shall not collect real names, phone numbers, location or identity documents from users; email is used only for sign-in and never shown to others. (Must)
privacy	NFR 12	Message content shall be encrypted at rest by Firestore's default encryption, with read access restricted by Security Rules to participants and reviewed moderator cases. (Should)
privacy	NFR 13	Chats shall be retained for a limited, configurable period (proposed 90 days) unless part of an open safety case. (Should)
Legal	NFR 14	Before any public launch: exclude under-16s, show a privacy notice, document a 48-hour breach procedure, assess National Cybersecurity Authority registration and the need for a data protection officer, and store personal data in Rwanda or under authorisation (Arts. 48 and 50). (Must)

5.4 Software quality attributes

Relative preference: safety and privacy come first, then usability, then performance and scalability

Requirement type	Req ID	Description
Usability	NFR 15	The interface shall use plain English suitable for secondary-school graduates and non-clinical wording. (Must)
Usability	NFR 16	Screens shall meet WCAG 2.1 AA contrast, be keyboard operable, and use tap targets of at least 44 px. (Should)

Usability	NFR 17	A signed-in user shall reach a chat request in three steps or fewer. (Must)
Reliability	NFR 18	Messages shall be saved before delivery so none is lost on failure; target availability of 95% during the pilot, excluding planned maintenance. (Should)
Auditability	NFR 19	Moderator and administrator actions and content access shall be logged with user, time and reason, and logs shall not be editable through the interface. (Should)
Maintainability	NFR 20	The message-screening rules shall be documented and version-controlled; code shall pass ESLint; a README shall cover set-up and deployment. (Must)
Testability	NFR 21	Automated tests shall run on every push, including against the Firebase Local Emulator Suite; the screening rules shall be tested with at least 40 labelled sample messages and their precision and recall reported. (Must)
Portability	NFR 22	Firebase project configuration shall be kept separate from source code, with distinct projects for development and the pilot. (Should)

5.5 Business rules

Rule	Description
BR 1	Users aged 16–17 are matched only with listeners who are 21 or older, approved, and cleared for minors.
BR 2	A listener must be approved by the administrator and pass training before accepting chats.
BR 3	Moderators may open chat content only through a case created by a flag or report, and every access is logged.
BR 4	Administrators and developers cannot read chats outside the moderation process.
BR 5	A user or listener may end a chat or block the other party at any time without giving a reason.
BR 6	Chats are confidential, except where there is serious risk of harm, when limited information may be shared with emergency services; users are told this at sign-up.

BR 7	A suspended account cannot chat until a moderator reinstates it.
BR 8	Tier 0 resources are always available without an account.

6. Appendix A: Glossary

Abbreviations and technical terms used in this document that aren't defined at first use.

Term	Definition
FR / NFR /BR	Functional Requirement, Non-Functional Requirement and Business Rule; the three requirement-identifier prefixes used in Sections 4 and 5, e.g. FR 4.2, NFR 6, BR 3.
MoSCow	The priority scale used on every requirement: Must have, Should have, Could have.
MVP	Minimum Viable Product; the smallest complete version of Spill (version 1.0) that delivers the four pilot objectives in Section 1.4.
Tier 0 / 1 / 2	Spill's access levels: Tier 0 is open to anyone with no account (resources page, crisis card); Tier 1 is a registered pseudonymous account (mood check-ins, listener chat); Tier 2, organisation-coded group spaces, is planned but out of scope for version 1.0.
Firestore	Firebase's real-time NoSQL database; chats, mood check-ins and profiles are stored here and sync to the browser within seconds.
Firebase Authentication	Firebase's service for sign-up, sign-in and session tokens; it hashes and stores passwords so the application never handles them in plain text.
Firestore Security rules	Server-enforced rules, written outside the application code, that decide who may read or write each piece of data; the mechanism behind role-based access and message screening
Firebase Local Emulator Suite	A local copy of Firestore and Authentication used to run automated tests without touching the live project.
ID token	A short-lived, signed credential Firebase Authentication issues after sign-in; the client attaches it to every Firestore request so Security Rules can check who is asking.

Crisis card	The always-available “Need help now” panel of verified local helplines and clinics, reachable in one tap from any page.

Appendix B: UML Diagrams

1. Use case Diagram

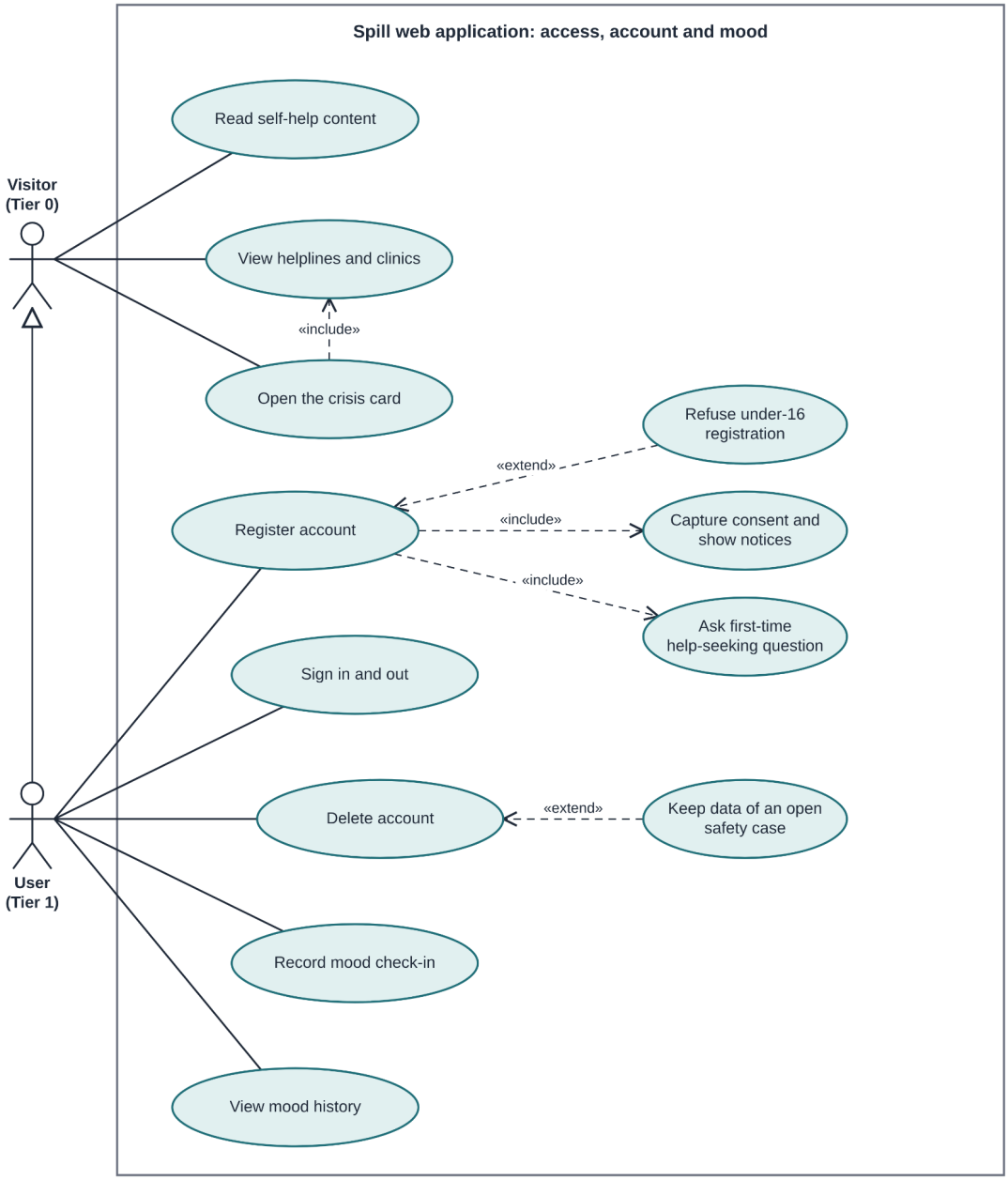


Figure B.1: *Use case diagram for the Visitor and the User*

Description & Flow:

The diagram shows what a person can do before and after registering. The actors are outside the system rectangle, and the ovals inside are the use cases.

- **Actors:** The Visitor uses Spill without an account. The User is a registered young person aged 16 to 25 and generalizes the Visitor, so a User can do everything a Visitor can.

- **Base use cases:** The Visitor can read self-help content, view helplines and clinics, and open the crisis card. The User can register an account, sign in and out, delete the account, record a mood check-in and view mood history.

- **Inputs and Outputs:** Inputs include age declarations, account credentials, and mood check-in values (1-5). Outputs include the created pseudonymous account, saved mood history, and displayed mental health resources.

- **Relationships**

Relationship	Type	Meaning
Open the crisis card to View helplines and clinics	«include»	The crisis card always shows the helplines.
Register account to Capture consent and show notices	«include»	Every registration records consent and shows the privacy notices.
Register account to Ask first-time help-seeking question	«include»	Every registration asks this one-time question.
Refuse under-16 registration to Register account	«extend»	Only if the age entered is under 16.
Keep data of an open safety case to Delete account	«extend»	Only if the account has an open safety case.

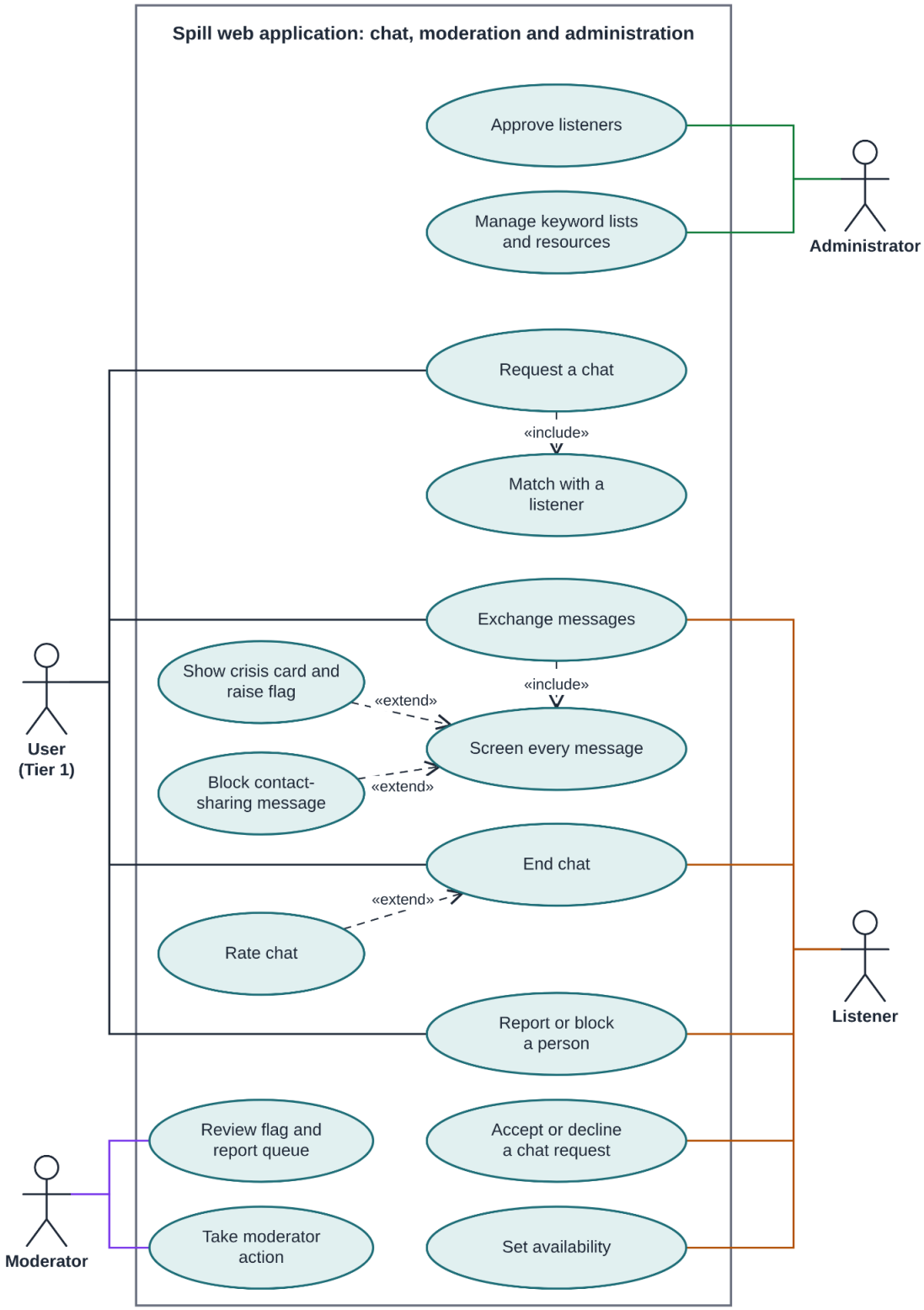


Figure B.2: *Use case diagram for chat and staff roles*

Description & Flow:

The diagram shows how Users and Listeners chat safely, and what the Moderator and Administrator do to support them. (Note: The line colours are only to help follow each actor and have no UML meaning).

- **Actors:** The User is the registered young person from Figure B.1. The Listener is an approved volunteer who supports Users. The Moderator reviews safety flags and reports. The Administrator manages listeners and content.

- **Base use cases:** The User can request a chat, exchange messages, end a chat, and report or block a person. The Listener can set availability, accept or decline a chat request, exchange messages, end a chat, and report or block a person. The Moderator can review the flag and report queue and take moderator action. The Administrator can approve listeners and manage keyword lists and resources.

- **Inputs and Outputs:** Inputs include chat requests, raw text messages, listener availability toggles, and moderator decisions. Outputs include matched chat sessions, delivered messages, generated safety flags, and updated system resources.

- **Relationships**

Relationship	Type	Meaning
Request a chat to Match with a listener	«include»	Every request goes through matching.
Exchange messages to Screen every message	«include»	Every message is screened before delivery.
Show crisis card and raise flag to Screen every message	«extend»	Only if crisis wording is found.
Block contact-sharing message to Screen every message	«extend»	Only if contact details are found.
Rate chat to End chat	«extend»	Only if the User chooses to rate the chat.

2. Class Diagram

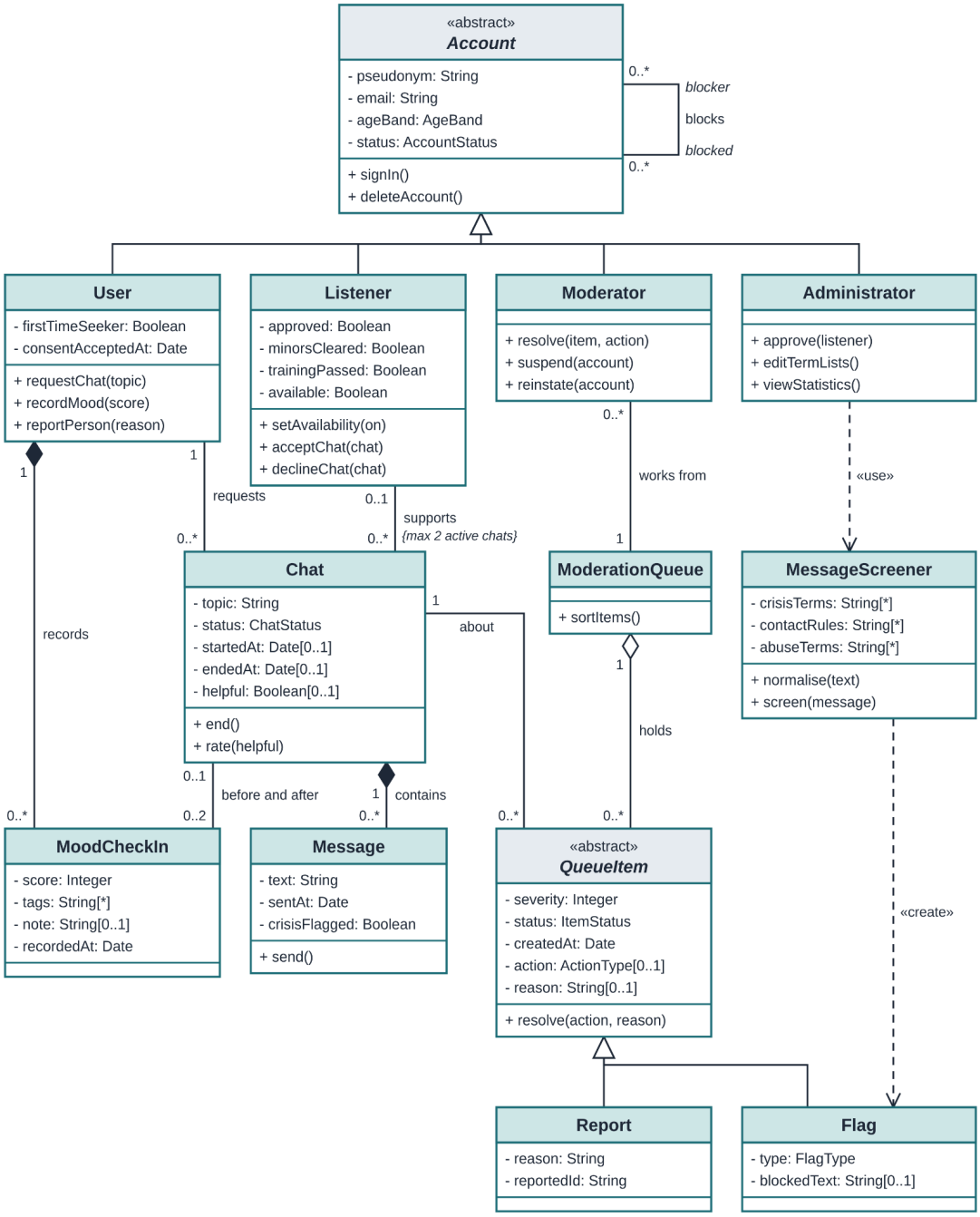


Figure B.3: *Class diagram of the Spill web application*

Description & Flow:

The diagram shows the main data and behaviour of Spill as classes. Each class lists its attributes (-, private) and operations (+, public), and the lines show how the classes relate. Abstract classes are in italics. A Visitor is not a class because a Visitor has no stored data.

- **Classes:** Account (abstract) holds what every signed-in person has, and User, Listener, Moderator and Administrator extend it with their own details and operations. Chat, Message and MoodCheckIn hold the conversations and mood entries. QueueItem (abstract), Flag, Report and ModerationQueue support moderation, and MessageScreener holds the rules used to screen messages.

- **Inputs and Outputs:** Inputs are the values users and staff enter, such as topics, messages, mood scores and moderator actions. Outputs are the stored records: chats, messages, mood entries, flags, reports and queue items.

- **Relationships**

Relationship	Type	Meaning
User, Listener, Moderator, Administrator to Account	Inheritance	Every role is an account and inherits the pseudonym, email, status and sign-in.
Flag, Report to QueueItem	Inheritance	Both appear in one moderator queue with a severity and status.
Account blocks Account (0..* to 0..*)	Association	A user or listener can block another account (FR 6.6).
User requests Chat (1 to 0..*)	Association	Only users start chats (FR 4.1, 4.4).
Listener supports Chat (0..1 to 0..*)	Association	A chat has one listener once accepted, with at most 2 active chats per listener (FR 5.5).
Chat to MoodCheckIn (0..1 to 0..2)	Association	Mood is recorded before and after a chat (FR 3.2).
Moderator works from ModerationQueue	Association	Moderators work from one queue (FR 6.7).
QueueItem about Chat (0..* to 1)	Association	Each flag or report concerns one chat.
User to MoodCheckIn	Composition	Mood entries are deleted with the account (FR 1.5).
Chat to Message	Composition	Messages exist only inside their chat.

ModerationQueue to QueueItem	Aggregation	The queue groups flags and reports, which exist independently of it.
MessageScreener «create» Flag	Dependency	Screening results in a flag being recorded when it finds crisis wording or blocked contact details (FR 6.2, 6.3).
Administrator «use» MessageScreener	Dependency	The Administrator edits the term lists the screener uses (FR 7.1).

3. Activity Diagram

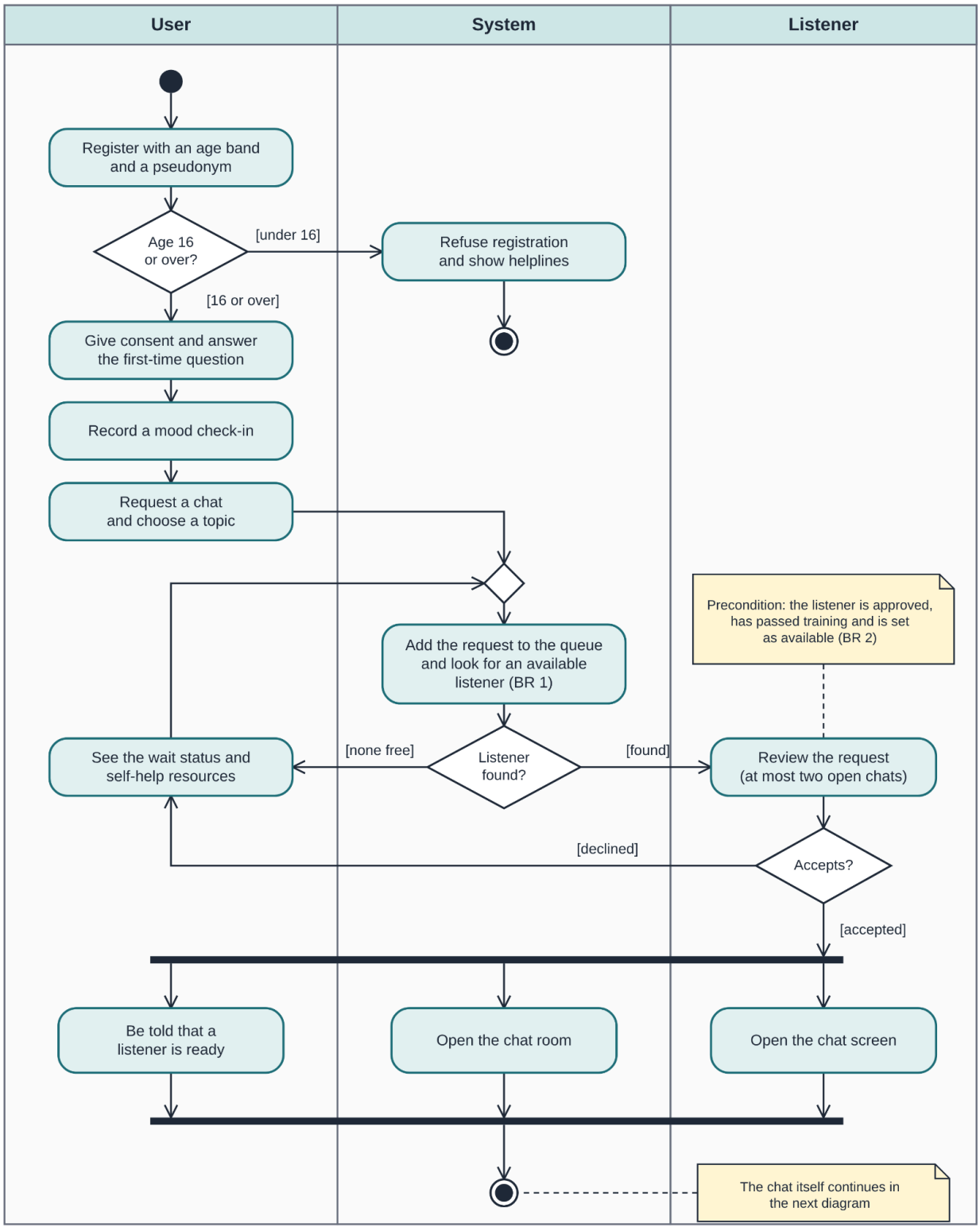


Figure B.4: *Activity diagram: Register and get matched*

Description & flow:

This diagram shows how a person goes from first visit to an open chat. It has three swimlanes: the User, the System and the Listener.

- **Inputs and Outputs:** Inputs include the user's age band, pseudonym, consent, first-time answer, mood score, and chat topic. Outputs include the system's decision state (refused registration, wait status) and the creation of an active chat room.

- **Flow and Activities:** The flow starts when the person registers with an age band and a pseudonym. A decision refuses anyone under 16 and ends the flow there. Everyone else gives consent, answers the first-time question, records a mood check-in and requests a chat with a topic. The System queues the request and looks for an available listener. Under BR 1, users aged 16–17 are matched only with cleared listeners. If none is free, or the listener declines, the user sees the wait status and self-help resources, and the request is retried. When a listener accepts, three actions run in parallel: the user is told a listener is ready, the chat room opens, and the listener's chat screen opens. A note records the precondition that the listener is approved, trained and available (BR 2).

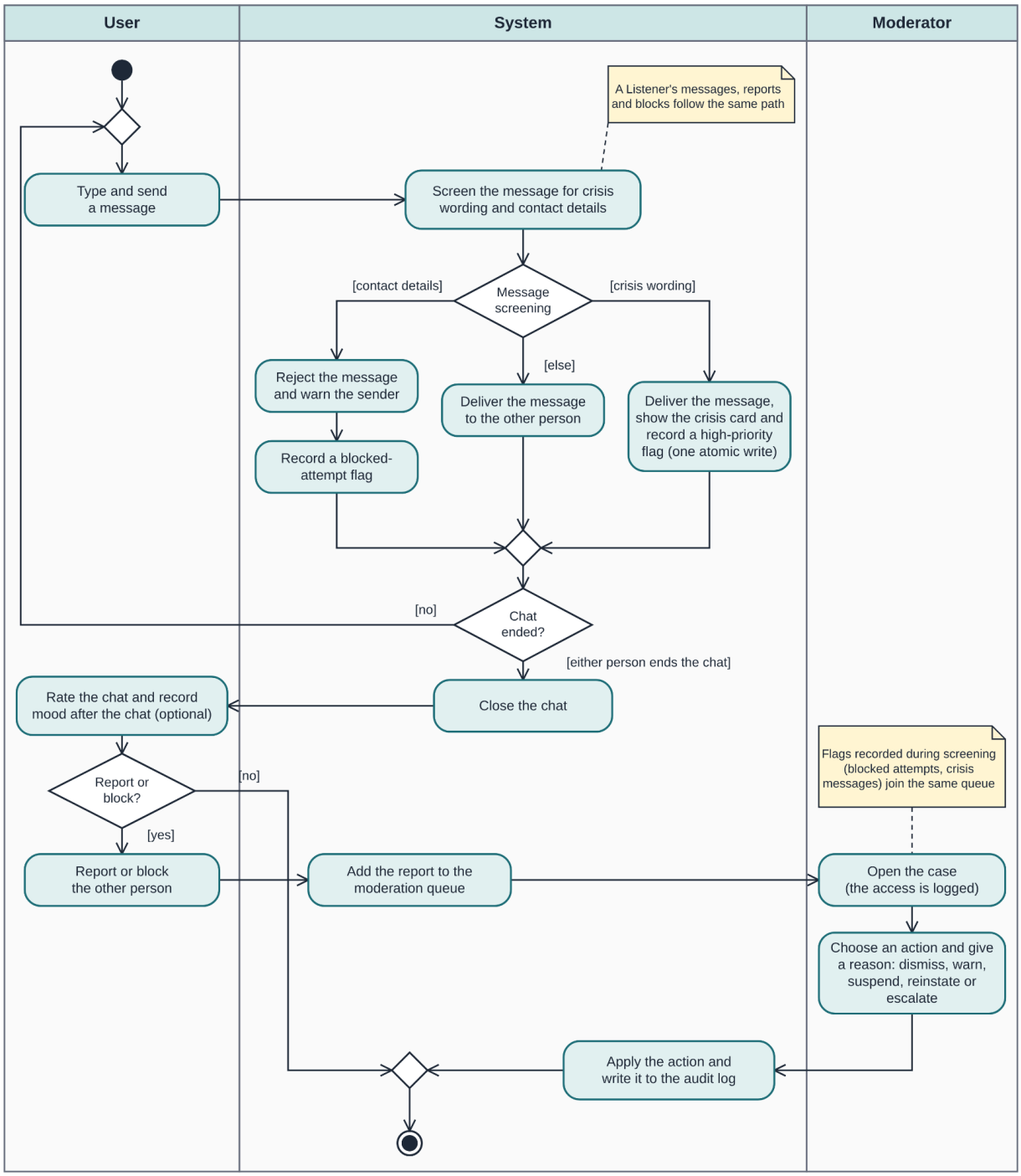


Figure B.5: *Activity diagram: Chat, screening and moderation*

Description & Flow:

This diagram shows how a chat works and how safety is handled. It has three swimlanes: the User, the System and the Moderator.

- **Inputs and Outputs:** Inputs include the typed text messages, chat ratings, post-chat mood scores, and the moderator's chosen action and reason. Outputs include delivered messages, sender warnings, generated safety flags/reports, and audit log entries.

- **Flow and Activities:** Each message the user sends is screened, and one of three things happens:

    - **Contact details:** the message is rejected, the sender is warned, and a blocked-attempt flag is recorded.
    - **Crisis wording:** the message is still delivered, the crisis card is shown, and a high-priority flag is recorded, all as one atomic write.
    - **Anything else:** the message is delivered.

The loop repeats until either person ends the chat. The user can then rate the chat and record a mood check-in. They can also report or block the other person, which adds a report to the moderation queue. The Moderator opens the case (the access is logged), chooses an action with a reason (dismiss, warn, suspend, reinstate or escalate), and the System applies it and writes it to the audit log. A note shows that flags recorded during screening join the same queue. A Listener's messages follow the same path.

4. Deployment Diagram

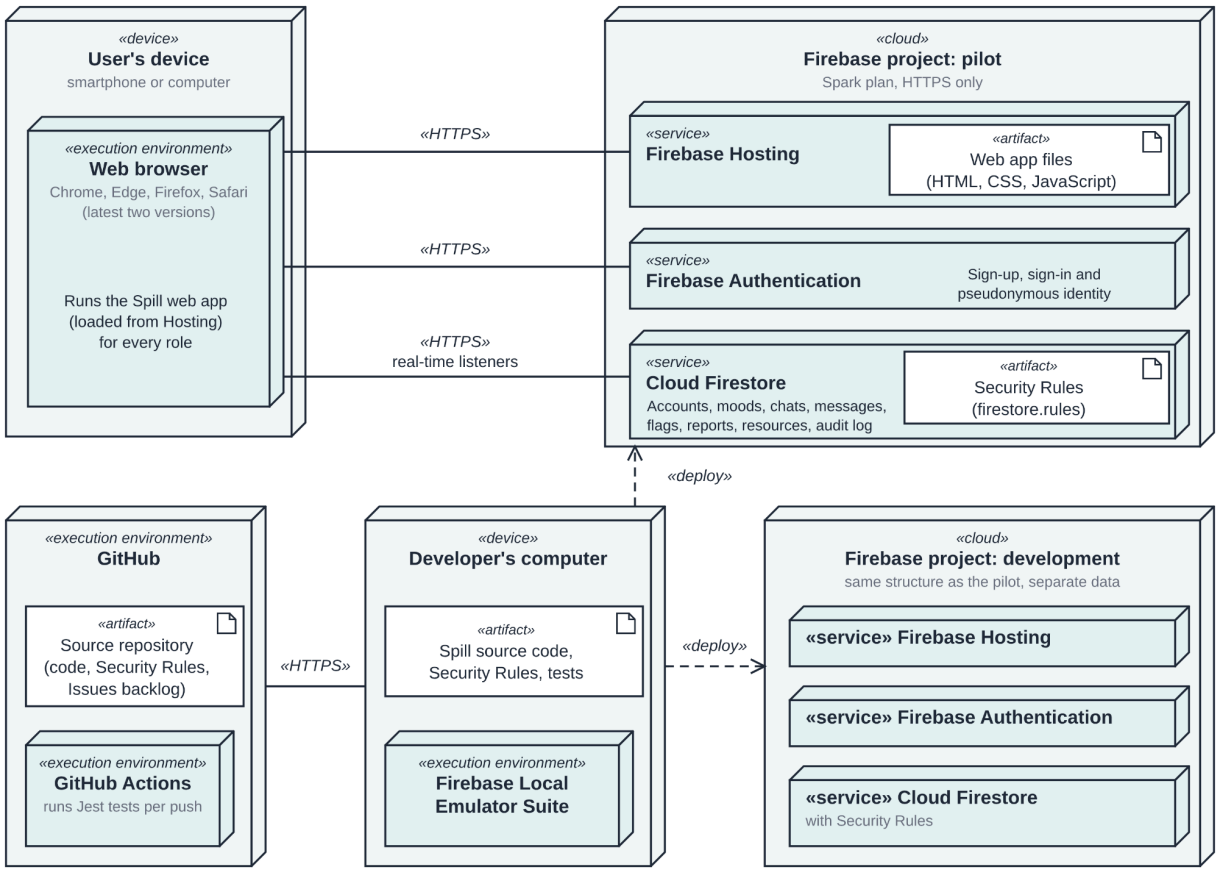


Figure B.6: *Deployment diagram of the Spill web application*

Description & Flow:

The diagram shows where each part of Spill runs and how the parts connect. The boxes are nodes, the page icons are artifacts, and the labels in « » name each node or link type.

- **Nodes:** the User's device runs a web browser that runs the web app. The Firebase pilot project has three services: Hosting, Authentication and Cloud Firestore. The Firebase development project has the same structure with separate data (NFR 22). The Developer's computer runs the Firebase Local Emulator Suite, and GitHub runs GitHub Actions.

- **Artifacts:** the web app files sit on Hosting, and the Security Rules sit on Firestore. The source code sits on the Developer's computer and in the GitHub repository.

- **Inputs and Outputs:** inputs include page requests, sign-ins, chat, mood and flag data, and pushed code. Outputs include the served web app, signed-in sessions, real-time updates, test results and deployed releases.

- **Relationships**

Connection	Type	Meaning
Browser to Hosting	«HTTPS»	Loads the web app.
Browser to Authentication	«HTTPS»	Signs the user up and in.
Browser to Firestore	«HTTPS»,real-time listeners	Reads and writes data, checked by Security Rules. No separate application server.
Developer's computer to GitHub	«HTTPS»	Pushes code, and GitHub Actions runs the Jest tests.
Developer's computer to the pilot and development projects	«deploy»	Releases the app to each Firebase project.